Universal regular amalgamation from a witness

Human Compression project

We recover the conditional existence assertion in Section 4 of the fixed Schulte–Weiss survey. Its premise is that an amalgamation already exists; the argument does not decide which local pairs have an amalgamation. We retain the standard end criterion for a string C-group and the correspondence between such marked groups and regular polytopes. Both are the same attributed inputs used by our acquired chamber theory. The construction below is standard universal-presentation reasoning, written out to keep its existence premise visible.

Theorem 1. *Let K, L be regular polytopes of rank $n \geq 2$. Suppose a polytope P of rank $n+1$ has all facets isomorphic to K and all vertex figures isomorphic to L. Then there is a regular polytope U with those local types which covers every polytope with those local types, after choosing base flags. Its group is obtained from the universal string group by imposing all the marked facet and vertex-figure relations.*

Proof. Let

$$W = \langle s_0, \ldots, s_n \mid s_i^2 = 1, (s_i s_j)^2 = 1 \ (|i-j| > 1) \rangle.$$

For $I \subseteq \{0, \ldots, n\}$ write $W_I = \langle s_i : i \in I \rangle$. Put $A = \{0, \ldots, n-1\}$, $B = \{1, \ldots, n\}$ and $C = A \cap B$. The standard markings give onto homomorphisms $W_A \to \mathrm{Aut}(K)$ and $W_B \to \mathrm{Aut}(L)$, with kernels N_A, N_B. Let N be their normal closure in W, and put $G = W/N$. Use G_I for the subgroup on the indicated images of the marks.

The generators of W act by the corresponding adjacency operations on the flags of P. These operations are involutions, and nonconsecutive ones commute, so this is an action of W (with words applied in their written order). Every relation in N_A acts trivially on all flags: it stays inside a facet, where the adjacency action is the regular flag action of a copy of K. This assertion holds in every facet, since an isomorphism of regular polytopes can send the chosen flag to any flag. The same argument in vertex figures applies to N_B. Consequently the action factors through G.

Fix a base flag Φ of P. The action of G_A on the flags containing the base facet is precisely the regular flag action of K. More explicitly, the quotient map induces $W_A/N_A \twoheadrightarrow G_A$, while the restriction of the action back to that facet has kernel exactly N_A in W_A. These two observations show that $G_A \cong W_A/N_A$ as marked groups. In particular G_A is a C-group and its action on this facet's flags is free. Similarly $G_B \cong W_B/N_B$, and its action on the flags containing the base vertex is free.

We claim $G_A \cap G_B = G_C$. If g belongs to the intersection, then Φ^g has both the base vertex and the base facet. The flags containing these two faces form the flag set of their section. They are connected by adjacencies of types $1, \ldots, n-1$. There is therefore $h \in G_C$ with $\Phi^h = \Phi^g$. Both h and g lie in G_A, whose action on the facet is free, so $g = h$. The reverse inclusion is immediate. The end criterion now proves that G is a string C-group. Its marked regular polytope U has facets K and vertex figures L by the exact end-group identifications just proved.

For any other polytope Q with those local types, the same local argument shows that N acts trivially on its flags. Choose a base flag Ψ. Identify the flags of U with G in such a way that adjacency of type i is right multiplication by s_i. The map

$$g \longmapsto \Psi^g$$

is well-defined, onto, and commutes with each adjacency operation. It induces a rank-preserving incidence map on faces: flags containing a fixed face of rank i form an orbit under adjacencies of all types except i, and their images therefore contain one fixed rank-i face of Q. This gives a face map independent of the chosen incident flag. Images of a flag remain a flag, and adjacent flags remain adjacent. Thus it is a polytope covering. This proves universality. □

The proof needs no regularity assumption on the witness P. In particular it includes a chiral witness with regular facets and vertex figures, as in the survey. It uses regularity of the two local types essentially: their marked relations must act trivially on every local flag, and the local action used to identify the middle intersection must be free. The oriented amalgamation assertion for chiral constituents has different input data, including relative enantiomorphic choices; it is retained separately with its source attribution.

The group presentation describes the universal object when the premise is satisfied. It does not show that every pair with matching intermediate sections has a witness, does not supply an order algorithm, and does not imply finiteness. Thus the theorem is the established conditional statement, not a solution of the survey's amalgamation or finiteness questions.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on polytopes, their groups, and realizations*, Period. Math. Hungar. 53 (2006), 231–255; fixed arXiv math/0608397v1, Section 4 and its attribution to P. McMullen and E. Schulte, *Abstract Regular Polytopes*. The standard end criterion and marked reconstruction are retained inputs, not new results of this note.